

US WEEKLY KICKSTART

Answers to 7 common client questions on potential impending mega IPOs

- **The potential for public launches of several large private companies has generated a wave of investor questions regarding IPOs.** The potential mega IPOs contribute to our forecast for a record \$160 billion in US IPO volume this year. Despite recent equity market volatility, our IPO Barometer continues to point to a conducive macro backdrop for new issuance. In this note we answer questions relating to the IPO outlook, the performance of recent IPOs, and the impact of potential large upcoming IPOs on the equity market.
- **Recent IPO performance has been mixed, with strong first-day rallies but below-average returns in subsequent months.** The average 2026 IPO has generated a first-day return of 19%, in line with the 30-year median. However, returns have deteriorated in following weeks, as reflected by the -1% YTD return of the GS Liquid IPO Index (GSX1LIPO). Consistent with the results of our long-term study of IPO returns, the best-performing recent IPOs have been characterized by elevated revenue growth and a near-term path to profitability.
- **The average recent IPO has traded particularly poorly ahead of lockup expiration, but the share price recovery afterward has been stronger than the historical average.** IPOs during the past decade have declined by an average of 4% in the weeks prior to the conclusion of the standard 180-day lockup window. This pattern has generally been more pronounced and more prolonged for recent offerings. Nonetheless, the average recent IPO has recovered to its initial closing pricing within three months following lockup expiration.
- **Historical parallels lend little support to investor concerns regarding the market impact of potential upcoming mega IPOs.** US equity mutual funds consistently increased cash balances ahead of the largest IPOs of the past few decades. Nonetheless, portfolio rotations around those episodes did not generate clear negative consequences for the S&P 500, the largest index constituents, or the Momentum factor.
- **The addition of large potential IPOs to major US benchmark indices would generate less selling pressure on current public equities than many investors fear, even accounting for potential index rule changes.** For example, if a hypothetical company with a \$100 billion float-adjusted market cap (\$1 trillion market cap and 10% float share) were to join the S&P 500, we estimate the consequent selling of current index constituents by active managers would equate to less than 5% of average daily volumes, and less than 10% including

Ben Snider

+1(212)357-1744 | ben.snider@gs.com
Goldman Sachs & Co. LLC

Ryan Hammond

+1(212)902-5625 | ryan.hammond@gs.com
Goldman Sachs & Co. LLC

Jenny Ma

+1(212)357-5775 | jenny.ma@gs.com
Goldman Sachs & Co. LLC

Daniel Chavez

+1(212)357-7657 | daniel.chavez@gs.com
Goldman Sachs & Co. LLC

Kartik Jayachandran

+1(212)855-7744 | kartik.jayachandran@gs.com
Goldman Sachs & Co. LLC

Christophe Sung

+1(212)902-3841 | christophe.sung@gs.com
Goldman Sachs & Co. LLC

passive funds.

For the exclusive use of SASCHA.SHA@SINOPAC.COM

Popular recent client questions on IPOs

The widely anticipated public debuts of several large private companies have sparked a number of recent client conversations on the topic of IPOs. Below we provide our perspective on the most common client questions on the topic:

1. How has IPO activity been recently, and what is the outlook for IPO activity going forward in 2026?
2. How have recent IPOs performed?
3. What characteristics have differentiated the best-performing recent IPOs?
4. How have IPOs traded around lockup expirations?
5. Are mutual funds likely to increase cash balances ahead of the upcoming IPOs?
6. How has the equity market traded around very large IPOs in the past?
7. What effect will index rule changes have on the market impact of large IPOs?

Q1: How has IPO activity been recently, and what is the outlook for IPO activity going forward in 2026?

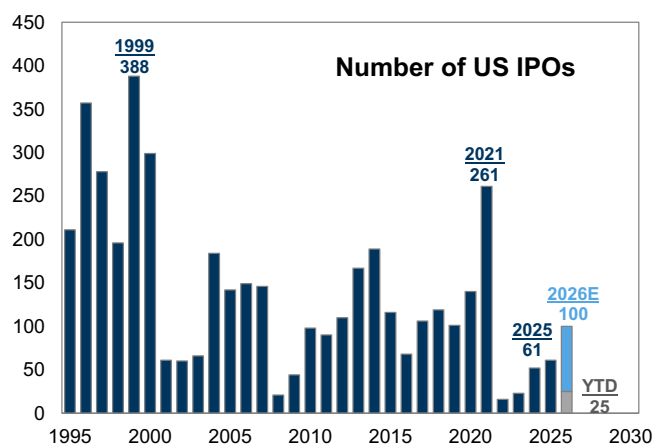
Thus far in 2026, there have been 25 IPOs greater than \$25 million in value, totaling \$14 billion in gross proceeds. This represents a nearly 80% increase in both the number and value of IPOs relative to this time last year. Roughly 40% of this year's IPOs have been Industrials companies compared with the historical annual average of 10% since 1995. In contrast, there have been no IPOs YTD in the Information Technology sector despite the sector representing 25% of IPOs since 1995.

We expect that roughly 100 IPOs totaling \$160 billion will come to market in 2026.

Our previous forecast was for 120 IPOs totaling \$160 billion. From a macro perspective, our GS IPO Barometer sits at 136 and continues to signal a conducive macro environment for IPOs. However, geopolitical uncertainty and recent equity market volatility suggest there will be fewer IPOs this year than we had previously expected. The 20% concentration of the IPO backlog since 2025 in Software is another headwind. From a micro perspective, several large private companies are expected to go public in coming quarters. Uncertainty around the timing and magnitude of these IPOs creates additional risks to our 2026 IPO forecast in both directions.

Exhibit 1: 25 companies have gone public so far this year

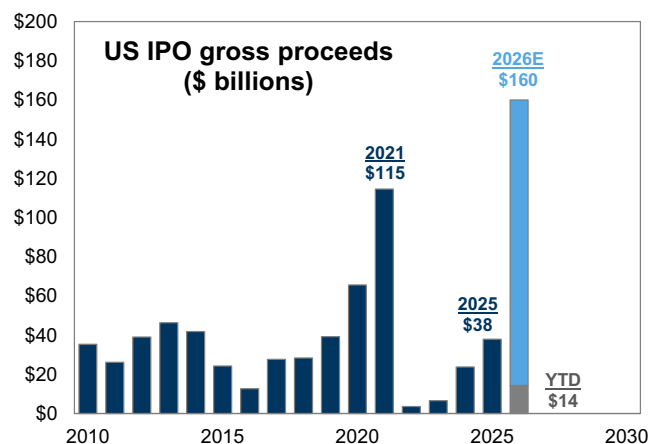
US deals greater than \$25 million in value



Source: FactSet, Goldman Sachs Global Investment Research

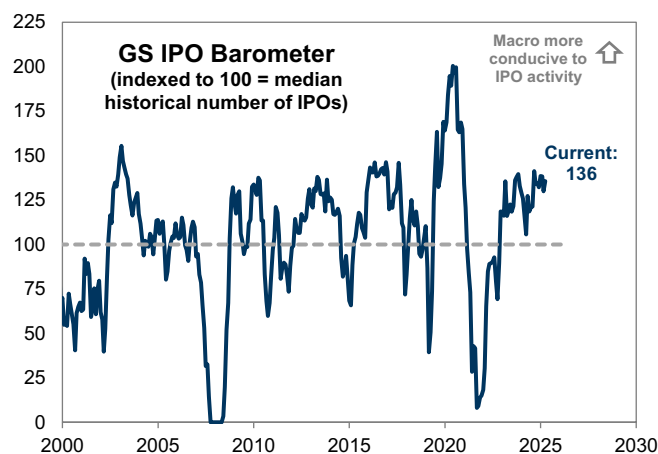
Exhibit 2: US IPOs have raised \$14 billion YTD

US deals greater than \$25 million in value



Source: FactSet, Goldman Sachs Global Investment Research

Going forward, we will update our IPO Barometer on a weekly basis, and we adjust the inputs slightly to allow for higher-frequency readings. See [here](#) for the original report discussing the Barometer, and [here](#) for data. Specifically, we replace the quarterly Conference Board survey of CEO confidence with the monthly CEO confidence survey from Chief Executive Magazine. This change would have had little impact on the historical series. The Barometer's other inputs, including interest rates and equity valuations, remain unchanged.

Exhibit 3: The macro environment remains conducive for IPOs

Source: Goldman Sachs Global Investment Research

Management commentary early in the Q1 2026 earnings season has signaled downside risk to the outlook for IPO activity but has also conveyed selective optimism for the second half of 2026.

- **Citigroup (C), April 14:** “You see selectivity—good quality deals getting done in IPOs and in debt capital markets. There is a bit of flight to quality in an environment like this—more momentum and activity in M&A and in high-grade debt, more caution

and moderation in high yield and IPOs, where quality is still happening but there is some risk-off.”

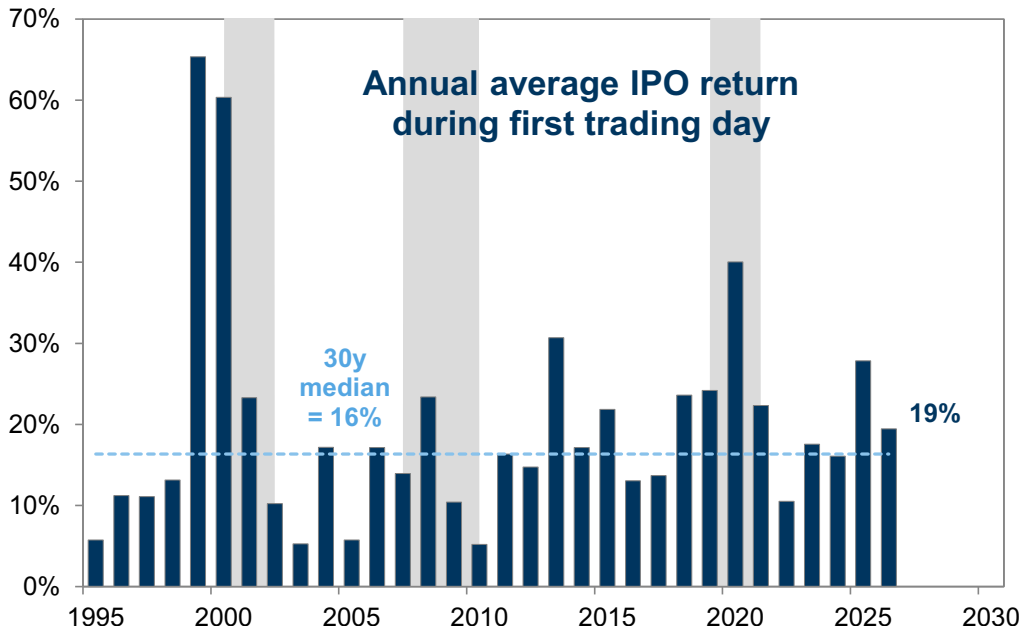
- **Wells Fargo & Co. (WFC), April 14:** “On the equity capital markets side, you have seen some delay in IPO activity in the latter part of the first quarter. Assuming some of the volatility subsides or stabilizes, you may see some of that start to come back. There is certainly a pipeline of companies waiting to go ... Overall, the pipeline and the expectation is still to see a pretty active rest of the year.”
- **Morgan Stanley (MS), April 15:** “I do think that not every company is going to be able to make it as an IPO in this environment. Some — there is going to be some selection ... And I think if we can get to a period now where we resume some of the narrative that we had going into 2026, which was a very strong one. I think what you will see is effectively the resumption of pipeline hitting the marketplace, whether it is through outright sales to other sponsors or likely to strategics or partial sales through IPOs, which is kind of a call that we collectively made going into the year. But I do think there is some selection.”

Supporting the signal from our IPO Barometer and the cautious optimism recently expressed by company managements, the IPO pipeline remains robust. 53 companies have filed to go public YTD, more than double the amount during the comparable period last year. Although the substantial weight of Software stocks in the IPO backlog remains a downside risk to IPO activity, none of the 19 companies filing for IPO since early April have been Software companies.

Q2: How have recent IPOs performed?

The average 2026 IPO has jumped by 19% on its first trading day, in line with the historical median return. The average 27% first-day return for the eight IPOs launched during the first three weeks of Q2 has been stronger than both the historical median and the 16% average first-day return for the IPOs during Q1, mirroring the rebound in the broader equity markets and signaling ongoing investor appetite for new issuance despite recent volatility.

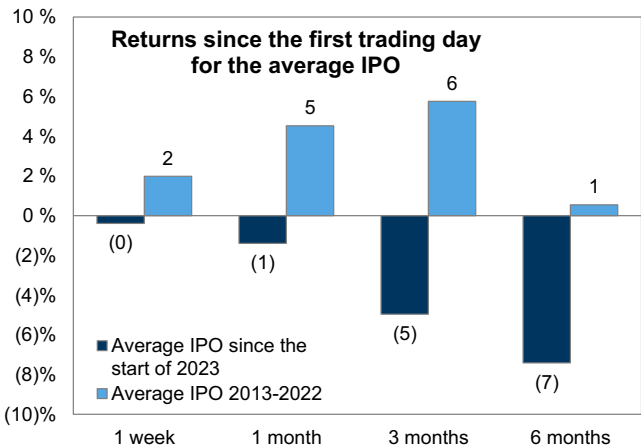
Exhibit 4: First day trading returns for the average IPO
US deals greater than \$25 million in value



Source: FactSet, Goldman Sachs Global Investment Research

However, strong initial returns have been short lived for recent IPOs. The average IPO since the start of 2023 has generated below-average returns during the first week, month, and quarter of trading relative to the pattern for IPOs during the past decade. As is generally the case, return dispersion among recent IPOs has been wide, but the return of the 90th percentile IPO since 2023 has also been slightly weaker than the comparable 90th percentile IPO in the prior decade.

Exhibit 5: Returns for the average recent IPO have been weaker than normal beyond the first trading day
161 US IPOs greater than \$25 million in value since 2023; 1283 IPOs during 2013-2022; bars reflect sample of IPOs with return data



Source: FactSet, Goldman Sachs Global Investment Research

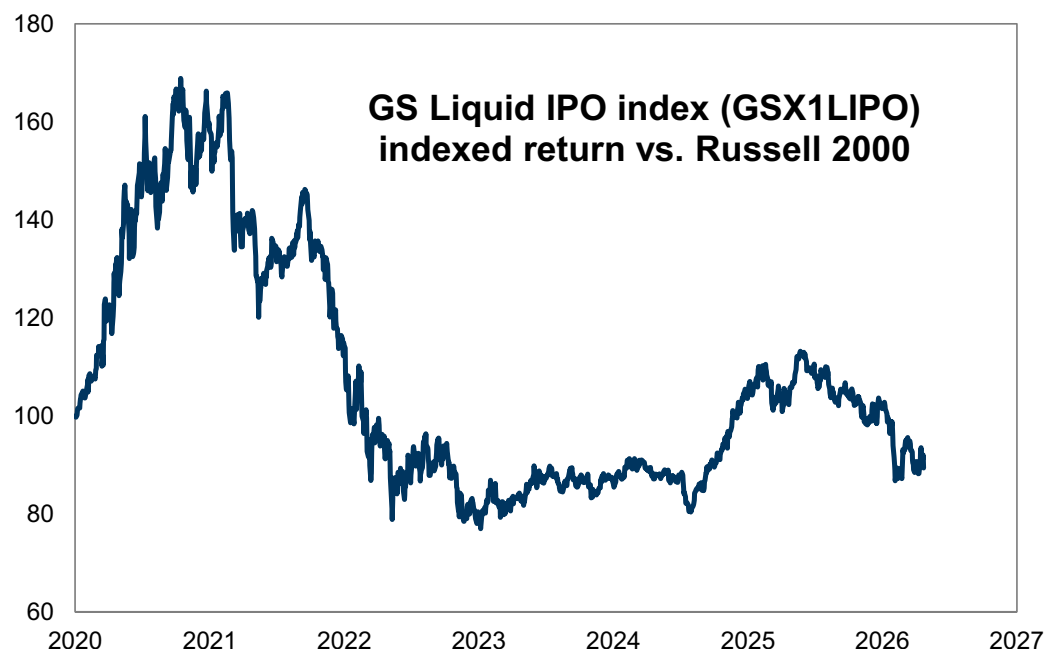
Exhibit 6: Returns for the 90th percentile recent IPO have lagged beyond the first trading day
161 US IPOs greater than \$25 million in value since 2023; 1283 IPOs during 2013-2022; bars reflect sample of IPOs with return data



Source: FactSet, Goldman Sachs Global Investment Research

The performance of the GS Liquid IPO Index (GSX1LIPO) reflects the below-average returns of recent IPOs. The index, which is composed on a rolling basis of the last 70 liquid US companies that were listed via IPO or direct listing, has returned -1% YTD vs. +12% for the Russell 2000.

Exhibit 7: The GS Liquid IPO Index (GSX1LIPO) has underperformed the Russell 2000



GSX1LIPO developed by GBM.

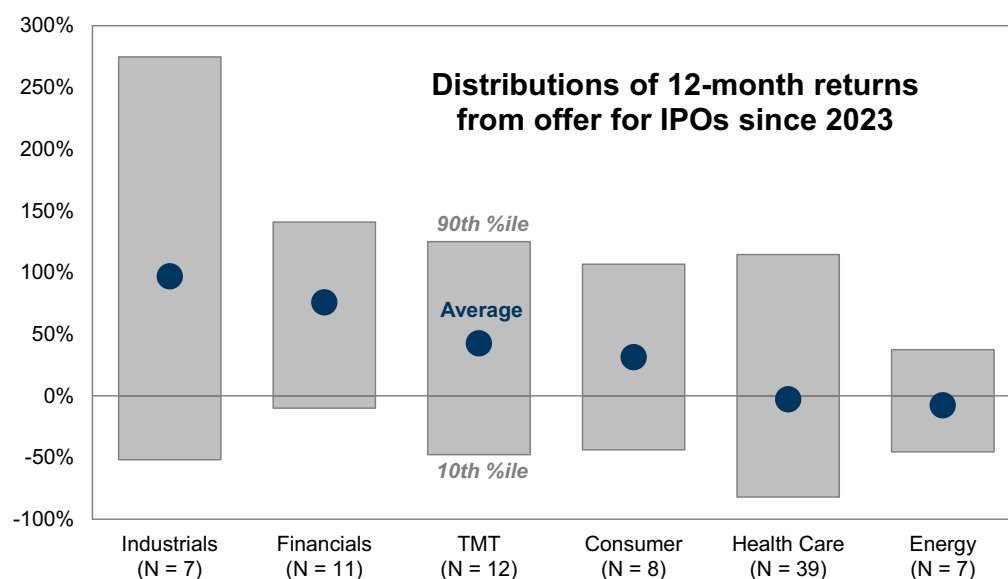
Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Q3: What characteristics have differentiated the best-performing recent IPOs?

At the sector level, Industrials IPOs have performed best on average during the last few years, although the distribution of returns has been extremely wide. In contrast, the average Health Care and Energy IPOs have delivered minimal 12-month post-IPO returns.

Exhibit 8: 1-year return distribution of recent IPOs by sector

US IPOs greater than \$25 million in value



Source: FactSet, Goldman Sachs Global Investment Research

From a fundamental perspective, the best-performing IPOs in recent years and recent decades have tended to be fast-growing companies with a short path to profitability. This was one of the conclusions of our study of nearly 5,000 IPOs during the past 25 years, and it has remained the case recently. Among 67 IPOs since 2023 with necessary return and consensus estimate data, the companies performing best during the 12 months post-IPO were those with consensus estimates indicating elevated sales growth and profitability within the 24 months post-IPO.

There has also been a relationship between valuation multiples at IPO and returns during the first 12 months of trading. Among IPOs launched since 2023, the average deal with a price/sales ratio below 2x returned 11% during the 12 months following its first day of trading, whereas the average deal with a price/sales multiple above 5x generated an average return of -5%.

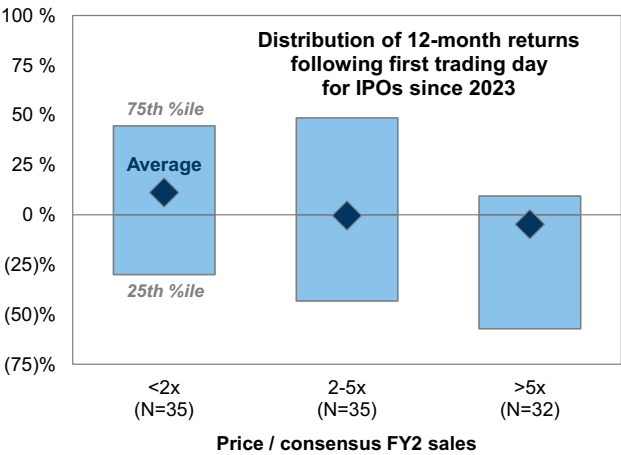
Exhibit 9: Sales growth and profitability differentiate the returns of recent IPOs

Average IPO 12-month return following first trading day			
		First year of profitability	
		Years 1 or 2	Later
Consensus 12m fwd sales growth	>20%	90 %	2 %
	5-20%	8	(29)
	<5%	(7)	(65)

67 US IPOs greater than \$25 million in value since 2023 with return and consensus estimate data; estimates reflect consensus expectations 12 months after IPO

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 10: Recent IPOs with lower valuations have tended to post better 12-month returns



Covers 102 IPOs greater than \$25 million in value since 2023 with return and consensus sales estimate data; price / sales calculated as (market cap at IPO offer price) / (consensus 2nd fiscal year sales as measured 1 month after IPO)

Source: FactSet, Goldman Sachs Global Investment Research

Q4: How have IPOs traded around lockup expirations?

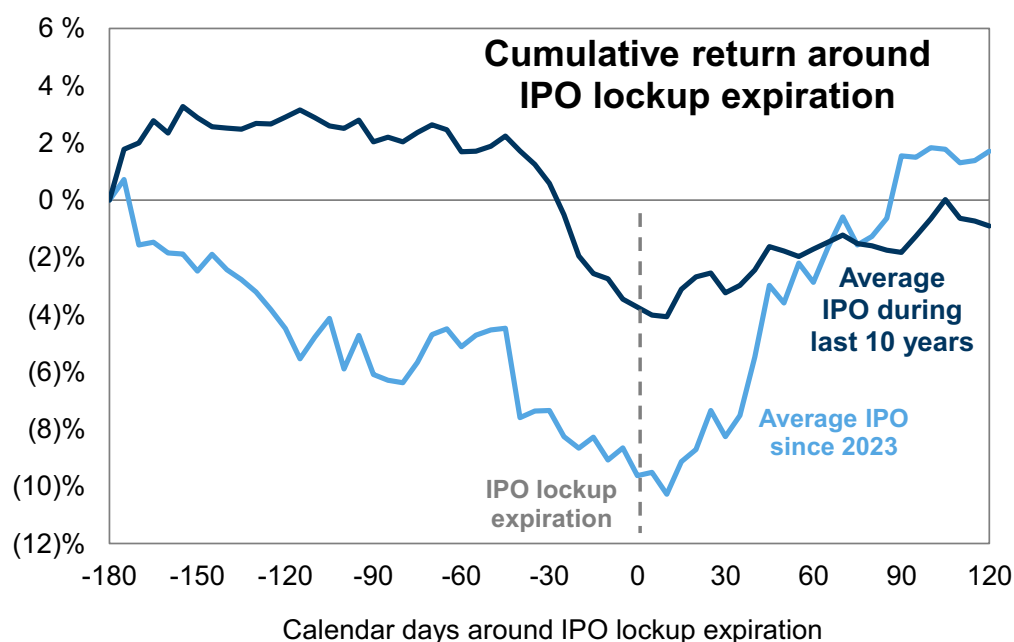
The expiration of an IPO’s lockup period has historically created a short-term performance headwind, on average, as new supply of shares entered the market.

Nearly all IPOs during the last decade have had a 180-day lockup period, during which most investors and insiders have been prohibited from selling their shares. The average IPO since 2017 has declined by about 5% during the month ahead of lockup expiration but has then stabilized and rebounded shortly thereafter.

The average IPO during the last few years has traded particularly poorly ahead of lockup expiration, but the recovery afterward has also been stronger than the historical average. The average IPO since 2023 has suffered a larger and more prolonged decline ahead of its lockup expiration than the average IPO of the past decade, falling by about 10% on average during the preceding 6 months. However, recent IPOs have followed the typical historical pattern of post-lockup rebound, rising on average above their first-day closing price within three subsequent months.

Exhibit 11: IPO returns around lockup expirations

US IPOs greater than \$25 million in value since 2017



Source: FactSet, Goldman Sachs Global Investment Research

Q5: Are mutual funds likely to increase cash balances ahead of the upcoming mega IPOs?

Few offerings in past decades have rivaled the reported size of the mega IPOs potentially coming to market in 2026. During the last 30 years, Visa (V), General Motors (GM), Alibaba (BABA), and Meta Platforms (META) rank as the largest IPOs by gross proceeds in both absolute terms and scaled to S&P 500 market cap at the time of IPO. The amount of capital raised by each of these four companies at IPO equated to about 15 basis points of aggregate S&P 500 market cap at the time. In today's terms, this would scale to a raise of roughly \$100 billion, similar to the hypothetical IPO of a company with a total market capitalization of \$2 trillion selling 5% of shares at launch.

Exhibit 12: Largest IPOs for companies trading on US exchanges since 1995 measured by gross proceeds as a share of S&P 500 market cap at time of IPO

Name	Ticker	IPO date	Size at time of IPO			
			Full market cap		Gross proceeds	
			Billions	Share of SPX cap	Billions	Share of SPX cap
Visa, Inc.	V	2008	\$42	37 bp	\$20	17 bp
General Motors Co.	GM	2010	50	46	18	17
Alibaba Group Holding Ltd.	BABA	2014	168	94	25	14
Meta Platforms, Inc.	META	2012	81	69	16	14

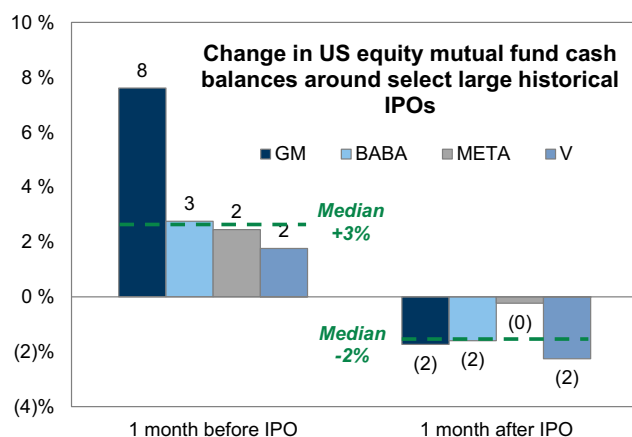
Source: FactSet, Goldman Sachs Global Investment Research

Mutual funds appear to strategically manage their cash positions around mega

IPOs. US equity mutual funds increased their cash balances ahead of each of the four largest IPOs of the past few decades. Cash balances rose by a median of 3% in the month prior to each of those IPOs but then declined by a median of 2% during the subsequent month.

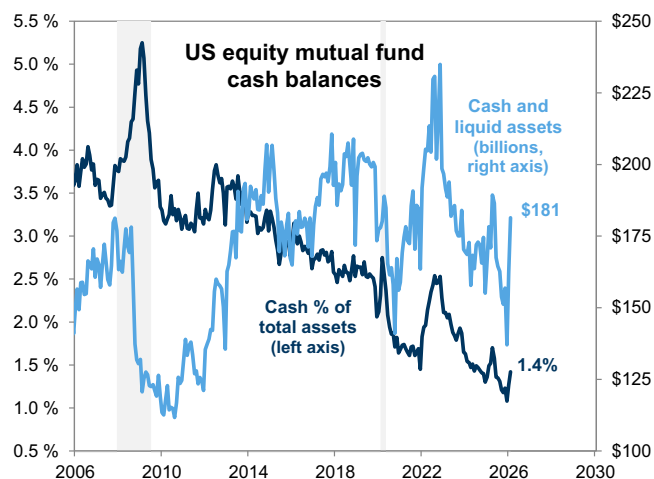
Relatively low mutual fund cash allocations today suggest funds will repeat the historical pattern and increase cash balances ahead of upcoming large IPOs. US equity mutual funds currently hold 1.4% of assets in cash, which ranks in the 5th percentile of the past 20 years. However, those cash allocations have risen since the recent lows in December 2025, and in dollar terms current balances of \$181 billion register in the middle of the range of the last several years.

Exhibit 13: Mutual funds have increased cash positions ahead of the largest IPOs in the past



Source: ICI, Goldman Sachs Global Investment Research

Exhibit 14: US equity mutual fund cash balances as of February 2026



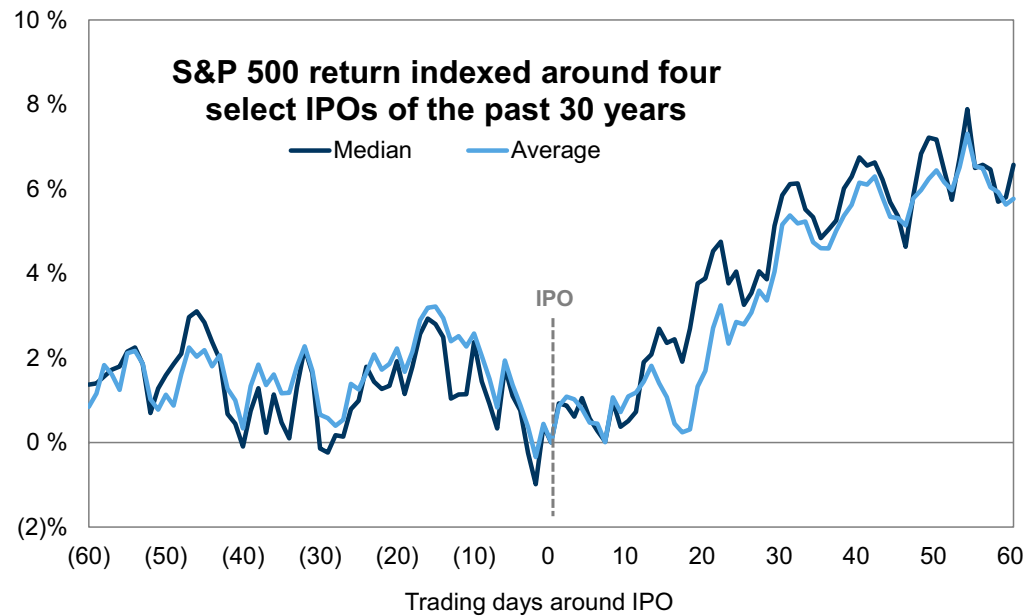
Source: ICI, Goldman Sachs Global Investment Research

Q6: How has the equity market traded around very large IPOs in the past?

The S&P 500 stalled immediately ahead of the largest IPOs in recent decades, but that dynamic appears more reflective of macro headwinds than of IPO-related supply pressures. On average, the S&P 500 declined by 1% in the month ahead of the four largest IPOs of the last 30 years ([Exhibit 12](#)). The equity market then rallied by an average of 5% during the subsequent month. However, US high yield credit performed similarly to equities around each of these IPOs, suggesting the path of the equity market was driven more by the macroeconomic backdrop than by IPO dynamics.

Exhibit 15: S&P 500 returns around large IPOs depend on the macro environment

select IPOs = BABA, GM, META, V

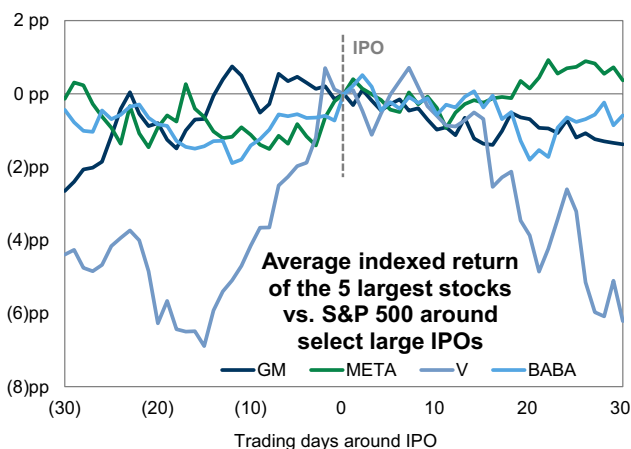


Source: FactSet, Goldman Sachs Global Investment Research

History shows no evidence of elevated selling pressure on the largest stocks ahead of major IPOs. The experiences of the BABA, GM, and META IPOs show the five largest S&P 500 stocks at each time trading roughly in line with the overall index, both before and after the IPOs. The exception was V, where the largest stocks outperformed ahead of the IPO, but that episode took place in 2008 during the financial crisis. Similarly, there has been no pattern of Momentum factor reversal around the largest IPOs in the past that would reflect investors selling popular holdings to make room in portfolios for the new issues.

Exhibit 16: Little evidence of selling pressure from large IPOs on the largest stocks

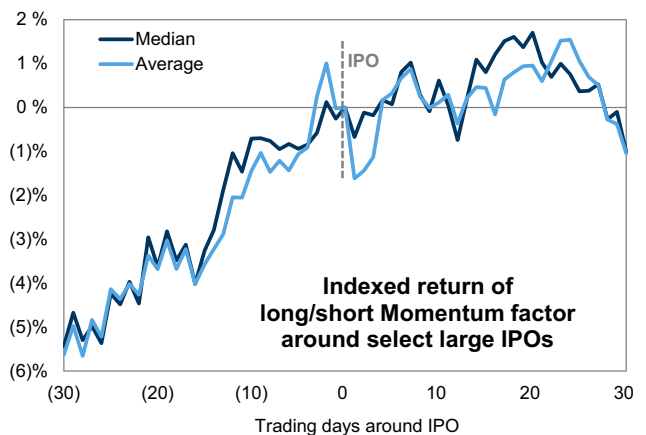
select IPOs = BABA, GM, META, V



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 17: Little evidence of selling pressure from large IPOs on past winners

select IPOs = BABA, GM, META, V



Source: FactSet, Goldman Sachs Global Investment Research

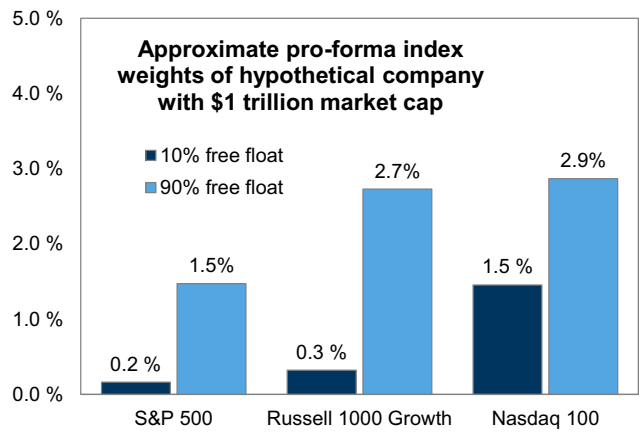
Q7: What effect will index rule changes have on the market impact of large IPOs?

The potential for mega IPOs this year has prompted Nasdaq to change its rules for index inclusion and weightings, and some other index providers have considered similar changes. Effective May 1st, Nasdaq will implement changes to the Nasdaq-100 Index that will fast-track inclusion for some large companies and boost the index weights of some securities with low float. FTSE Russell has launched a public consultation seeking feedback regarding similar potential changes. News reports have suggested that S&P Dow Jones Indices is considering similar changes, but the most recent blog post on the topic from the provider in March suggests no changes will be made to S&P 500 inclusion rules without holding a public consultation period, and none has been announced so far. We discussed the potential changes and their potential consequences last month.

Fast-tracked mega IPOs would carry small weights in the major benchmark US indices at first, but would grow substantially once more shares are floated. A hypothetical company with a total market capitalization of \$1 trillion and 10% of free float would have a weight of less than 0.5% in the S&P 500 and Russell 1000 Growth indices and a weight of less than 2% in the Nasdaq-100. However, holding all else equal, the company's weight across these indices would grow to 2-3% once its free float reached 90%.

The hypothetical inclusion of a large IPO in major benchmark indices soon after launch would be unlikely to create significant selling pressure on current index constituents. Roughly \$18 trillion of fund AUM is benchmarked to the S&P 500 Index, and an additional \$5 trillion is benchmarked to the Russell 1000 Growth and Nasdaq-100. If funds benchmarked to the S&P 500 added a hypothetical IPO with \$1 trillion market cap and 10% free float to their portfolios at benchmark weight, the reallocation flows would create selling pressure on existing index constituents equating to about 5 bp of those constituents' collective market cap and less than 10% of their average daily trading volume.

Exhibit 18: Pro-forma index weights of a hypothetical large IPO entering US indices



Assumes all market cap is listed.

Source: Goldman Sachs Global Investment Research

Exhibit 19: Potential selling pressure on current constituents from the hypothetical addition of a large index constituent

Hypothetical selling pressure on current index constituents from funds in response to index inclusion of a \$100 billion float-adjusted company						
Index	Pro-forma index weight (%)	Estimated benchmarked fund AUM (\$ trillions)		Hypothetical selling pressure on current constituents		
		Active	Passive	\$ billions	Share of mkt cap (bp)	% of ADTV
S&P 500	0.2	8	10	30	5	8
Russell 1000 Growth	0.3	4	1	15	5	5
Nasdaq 100	1.5	0	1	9	3	4

We estimate \$6 billion of active mutual fund AUM benchmarked to the Nasdaq-100 Index.

Source: Goldman Sachs Global Investment Research

Largest US IPOs since 2025

Exhibit 20: Largest IPOs since 2025

US deals greater than \$25 million in value

Name	Ticker	Sector	IPO date	Size at time of IPO (\$ billions)		Return to date since IPO
				Market cap	Gross proceeds	
Venture Global, Inc.	VG	Energy	Jan-2025	\$60	\$1.8	(48)%
Medline, Inc.	MDLN	Health Care	Dec-2025	38	7.2	59
Figma, Inc.	FIG	Information Technology	Jul-2025	16	1.4	(48)
CoreWeave, Inc.	CRWV	Information Technology	Mar-2025	14	1.6	194
Madison Air Solutions Corp.	MAIR	Industrials	Apr-2026	13	2.2	27
SailPoint, Inc.	SAIL	Information Technology	Feb-2025	13	1.4	(50)
Fermi Inc.	FRMI	Real Estate	Oct-2025	12	0.8	(72)
Arxis, Inc.	ARXS	Industrials	Apr-2026	11	1.3	33
Chime Financial, Inc.	CHYM	Financials	Jun-2025	10	1.0	(16)
StubHub Holdings Incorporation	STUB	Communication Services	Sep-2025	9	0.8	(70)
Forgent Power Solutions, Inc.	FPS	Industrials	Feb-2026	8	1.7	28
BETA Technologies, Inc.	BETA	Industrials	Nov-2025	7	1.2	(55)
Smithfield Foods, Inc.	SFD	Consumer Staples	Jan-2025	7	0.6	46
Netskope, Inc.	NTSK	Information Technology	Sep-2025	7	1.0	(48)
Circle Internet Group, Inc.	CRCL	Information Technology	Jun-2025	7	1.2	223
Firefly Aerospace, Inc.	FLY	Industrials	Aug-2025	6	1.0	(14)
Navan, Inc.	NAVN	Consumer Discretionary	Oct-2025	6	0.9	(38)
NIQ Global Intelligence PLC	NIQ	Communication Services	Jul-2025	6	1.1	(47)
EquipmentShare.com, Inc.	EQPT	Industrials	Jan-2026	6	0.9	0
Caris Life Sciences, Inc.	CAI	Health Care	Jun-2025	6	0.5	(6)

Source: FactSet, Goldman Sachs Global Investment Research

Earnings season summary

Exhibit 21: S&P 500 1Q 2026 earnings results

as of April 24, 2026

S&P 500 EQUAL-WEIGHTED												
	Number of Companies			EARNINGS						REVENUE		
				Std Dev Surprises			Absolute Surprises		Avg 1Q Surprise	Std Dev Surprises		Avg 1Q Surprise
	Reported	Total	% of Co's	Positive	Negative	In-Line	Positive	Negative		Positive	Negative	
Information Technology	14	73	19%	86 %	0 %	14 %	100 %	0 %	9 %	79 %	0 %	4 %
Materials	5	26	19	80	0	20	80	20	24	40	0	4
Health Care	12	58	21	75	0	25	83	8	8	58	8	2
Consumer Staples	9	36	25	67	11	22	78	22	3	78	0	2
Energy	5	22	23	60	0	40	80	20	10	40	0	3
Industrials	29	79	37	55	0	45	76	10	10	48	3	2
Financials	41	76	54	54	5	39	85	12	5	37	22	1
Communication Services	4	20	20	50	0	50	75	25	16	75	0	1
Real Estate	4	31	13	50	0	50	50	50	11	75	0	1
Utilities	3	31	10	33	0	67	67	33	12	33	0	2
Consumer Discretionary	13	48	27	31	15	54	62	31	3	23	31	(0)
S&P 500	139	500	28%	58 %	4 %	37 %	80 %	15 %	8 %	49 %	11 %	2 %

Comparative Data (full earnings season)

4Q 2025	55 %	12 %	33 %	73 %	23 %	11 %	51 %	13 %	2 %
3Q 2025	66	8	26	82	15	10	52	9	2
2Q 2025	60	9	31	78	17	9	53	7	2
1Q 2025	54	11	35	73	22	7	36	15	1

S&P 500 CAP-WEIGHTED												
	Weight of Companies			EARNINGS						REVENUE		
				Std Dev Surprises			Absolute Surprises		Avg 1Q Surprise	Std Dev Surprises		Avg 1Q Surprise
	Reported	Total	% of Co's	Positive	Negative	In-Line	Positive	Negative		Positive	Negative	
Information Technology	4	35	11%	94 %	0 %	6 %	100 %	0 %	16 %	83 %	0 %	8 %
Materials	0	2	24	89	0	11	89	11	26	73	0	7
Communication Services	1	11	11	81	0	19	97	3	36	97	0	1
Financials	7	12	59	70	2	28	94	5	8	60	15	2
Industrials	4	9	46	65	0	35	88	6	15	43	5	2
Consumer Staples	2	5	45	65	1	33	98	2	3	66	0	2
Real Estate	0	2	24	63	0	37	63	37	15	52	0	(1)
Energy	0	3	13	57	0	43	71	29	9	35	0	3
Health Care	3	9	35	57	0	43	87	4	6	38	6	2
Utilities	0	2	19	14	0	86	89	11	16	14	0	(4)
Consumer Discretionary	2	10	23	8	3	89	92	6	7	3	9	(1)
S&P 500	26	100	26%	65 %	1 %	34 %	93 %	5 %	11 %	54 %	7 %	3 %

Source: FactSet, Goldman Sachs Global Investment Research

S&P 500 earnings and return forecasts

Pricing as of April 23, 2026, unless otherwise noted.

Exhibit 22: Goldman Sachs top-down S&P 500 forecasts



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 23: Goldman Sachs and consensus forecasts for S&P 500 earnings and returns

	Goldman Sachs forecast	Bottom-up analyst consensus	Top-down strategist consensus
S&P 500 EPS			
2025	\$275	\$275	\$275
Y/Y growth	12 %	12 %	12 %
2026	\$309	\$327	\$310
Y/Y growth	12 %	19 %	13 %
2027	\$342	\$379	\$349
Y/Y growth	10 %	16 %	13 %
S&P 500 P/E			
Current on NTM EPS	22x	21x	22x
S&P 500 price target			
YE 2026	7600	NA	7500
Return to target	7 %		6 %
12-month	7600	8200	NA
Return to target	7 %	15 %	

Source: Bloomberg, FactSet, Goldman Sachs Global Investment Research

Biggest stock movers this week

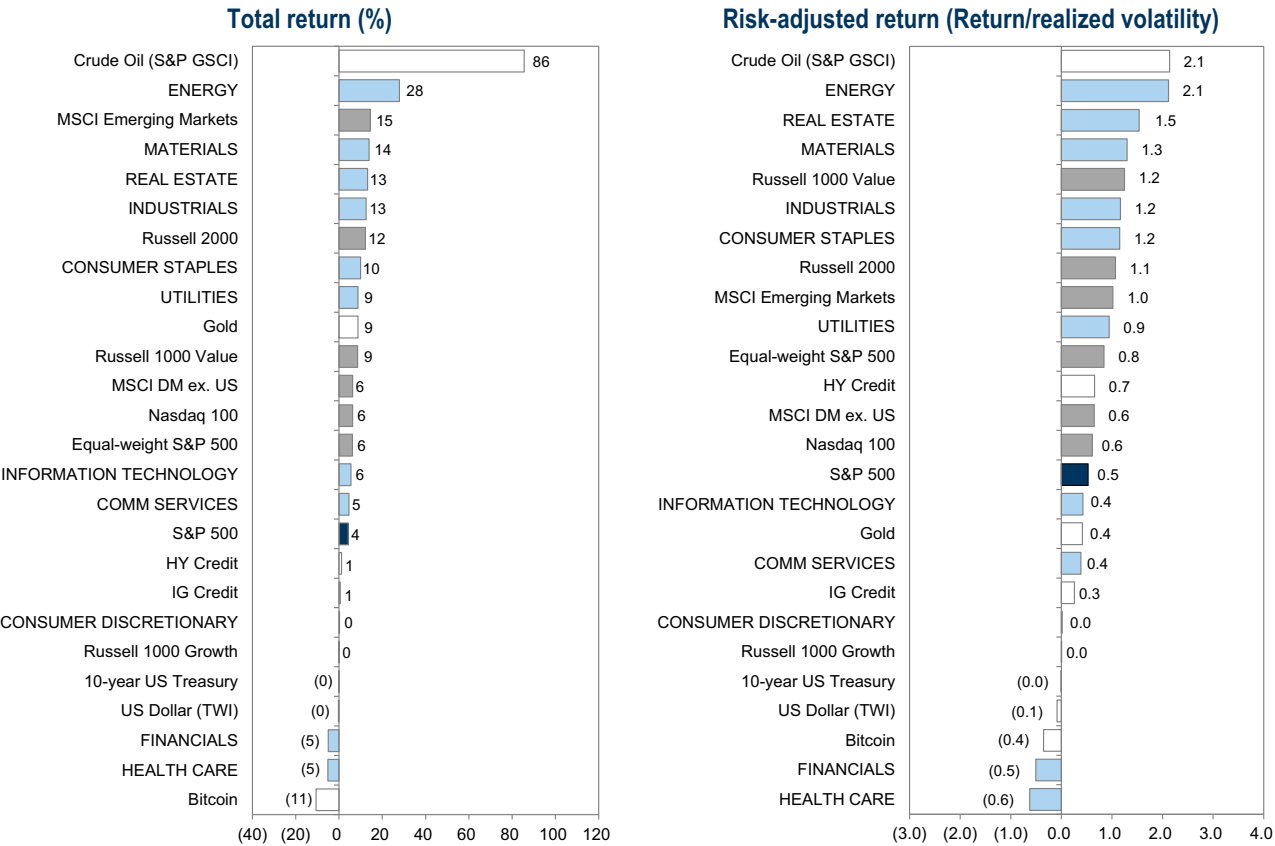
Exhibit 24: S&P 500 stocks with the largest moves this week as of April 24, 2026

Company	Ticker	Industry group	Total return				Market cap (bn)	Consensus	Fwd 12m
			This week	3m	6m	12m		2026 EPS growth	P/E
Top 10 returns this week									
Advanced Micro Devices	AMD	Semiconductors & Semi Equip	25 %	18 %	30 %	238 %	\$498	62 %	52x
United Rentals Inc.	URI	Capital Goods	22	8	8	69	62	8	20
Texas Instruments	TXN	Semiconductors & Semi Equip	21	47	66	91	257	37	35
Intel Corp.	INTC	Semiconductors & Semi Equip	20	48	75	224	335	42	80
ON Semiconductor	ON	Semiconductors & Semi Equip	19	58	89	167	38	21	34
Baker Hughes	BKR	Energy	15	20	33	83	64	(6)	27
GE Vernova Inc.	GEV	Capital Goods	15	75	93	243	309	110	65
Microchip Technology	MCHP	Semiconductors & Semi Equip	14	22	41	123	49	109	37
Steel Dynamics Inc.	STLD	Materials	13	24	50	86	33	78	14
NXP Semiconductors	NXPI	Semiconductors & Semi Equip	13	4	10	36	61	16	18
Bottom 10 returns this week									
Charter Communications	CHTR	Media & Entertainment	(24)%	26 %	(1)%	(28)%	\$30	20 %	4x
Tractor Supply	TSCO	Consumer Discretionary Retail	(18)	(30)	(32)	(23)	20	4	17
lululemon athletica inc.	LULU	Consumer Durables & Apparel	(14)	(26)	(21)	(47)	16	(8)	12
Northrop Grumman	NOC	Capital Goods	(14)	(12)	(2)	26	83	5	20
Lockheed Martin	LMT	Capital Goods	(13)	(10)	10	17	122	29	17
Freeport-McMoRan	FCX	Materials	(13)	2	50	77	88	64	20
TE Connectivity	TEL	Tech Hardware & Equipment	(13)	(2)	(6)	62	64	20	18
Equifax Inc.	EFX	Commercial & Professional Svcs	(12)	(16)	(24)	(29)	21	11	19
International Paper	IP	Materials	(12)	(21)	(30)	(25)	18	NM	22
Norwegian Cruise Line Hldgs	NCLH	Consumer Services	(12)	(12)	(22)	8	8	4	9
Median S&P 500 stock			(1)%	1 %	5 %	19 %	\$42	9 %	19x
S&P 500 Index			1	3	6	34	63,610	19	21

Source: FactSet, Goldman Sachs Global Investment Research

YTD absolute and risk-adjusted returns

Exhibit 25: YTD asset returns and return/volatility ratios



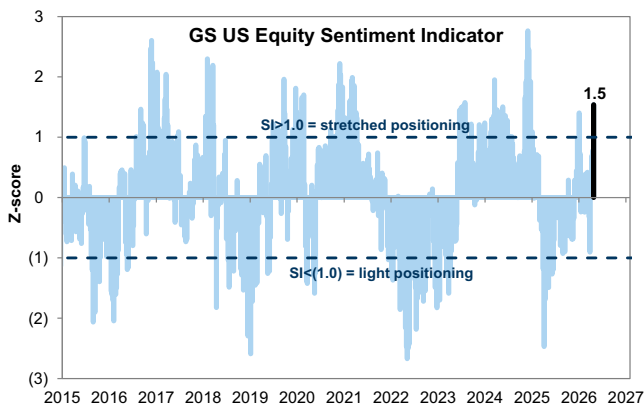
Crude oil represents S&P GSCI Crude Oil Index.

Source: FactSet, Haver Analytics, Goldman Sachs Global Investment Research

For the exclusive use of SASCHA.SHA@SINOPAC.COM

Sentiment and flows

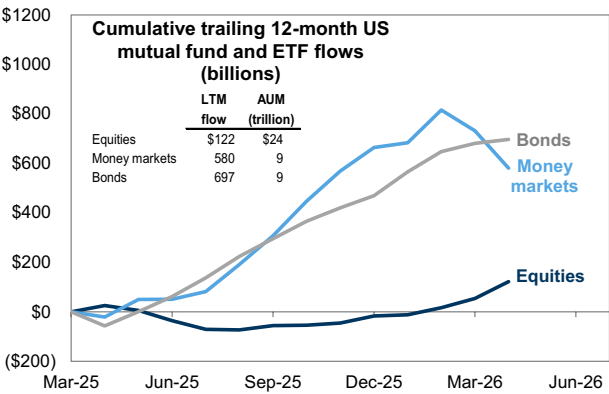
Exhibit 26: GS US Equity Sentiment Indicator of investor positioning as of April 24, 2026



The Sentiment Indicator combines 9 measures of positioning across institutional, retail, and foreign investors and has historically been a statistically significant signal for near-term S&P 500 returns.

Source: Goldman Sachs Global Investment Research

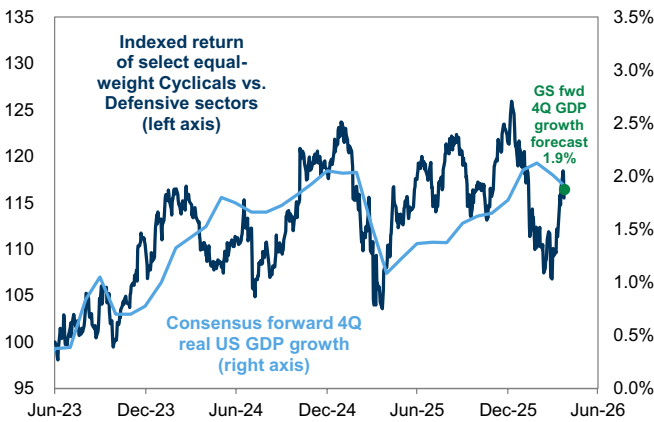
Exhibit 27: Recent mutual fund and ETF flows



Source: EPFR, Goldman Sachs Global Investment Research

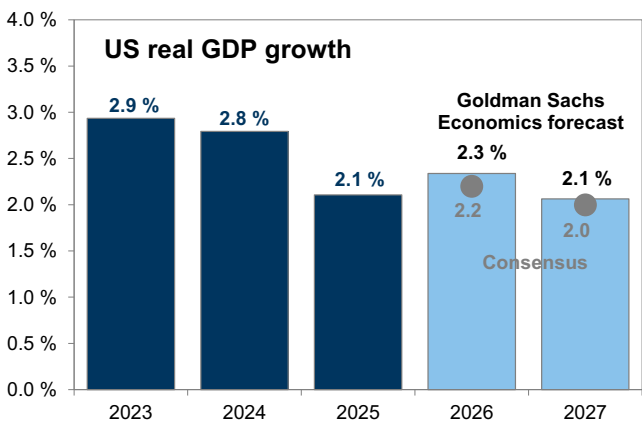
Economic growth

Exhibit 28: US equity market internal pricing of economic growth



Source: Bloomberg, Goldman Sachs Global Investment Research

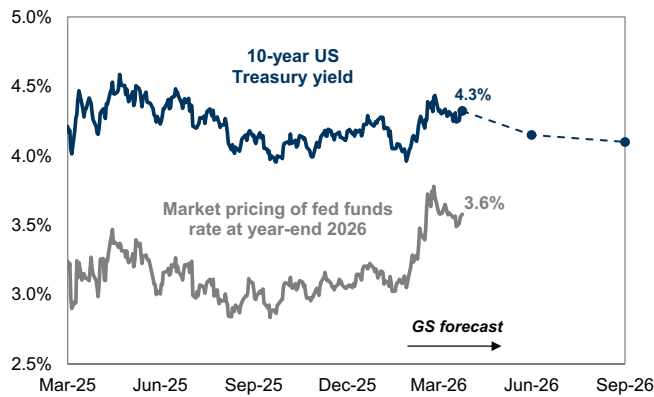
Exhibit 29: Goldman Sachs and consensus forecasts for US real GDP growth



Source: Bloomberg, Goldman Sachs Global Investment Research

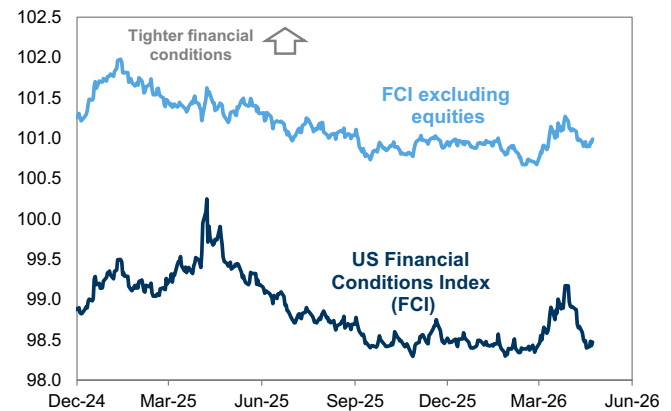
Interest rates and financial conditions

Exhibit 30: Market pricing of near-term and long-term interest rate outlook



Source: Goldman Sachs Global Investment Research

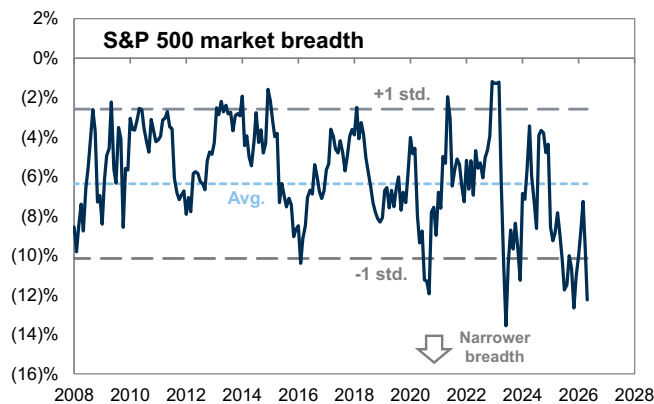
Exhibit 31: Goldman Sachs Financial Conditions Index



Source: Goldman Sachs Global Investment Research

Market breadth and concentration

Exhibit 32: S&P 500 52-week market breadth



Market breadth calculated as the difference between the distance of the aggregate S&P 500 Index from its 52-week high and the distance of the median S&P 500 constituent from its 52-week high. When the aggregate index is much closer to its 52-week high than is the median stock, market breadth is narrow.

Source: Goldman Sachs Global Investment Research

Exhibit 33: Concentration of S&P 500 market cap and earnings in the 10 largest index constituents

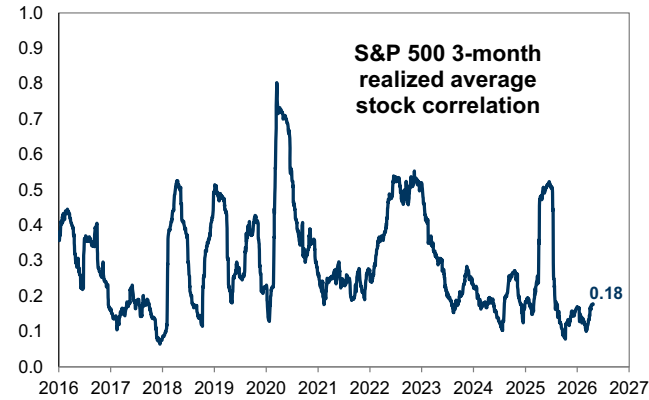


Earnings reflect consensus forward 12-month estimates.

Source: Compustat, IBES, FactSet, Goldman Sachs Global Investment Research

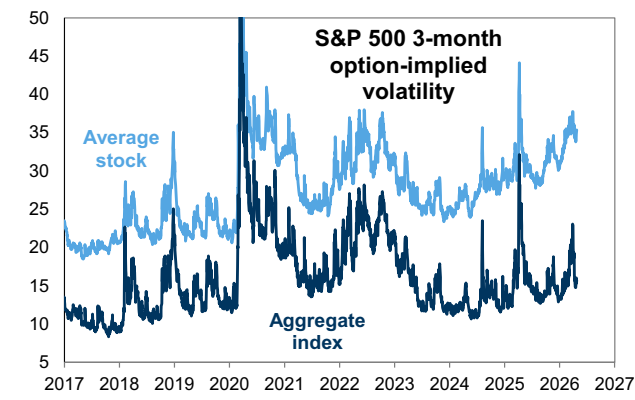
Correlation and volatility

Exhibit 34: S&P 500 realized average stock correlation



Source: Goldman Sachs Global Investment Research

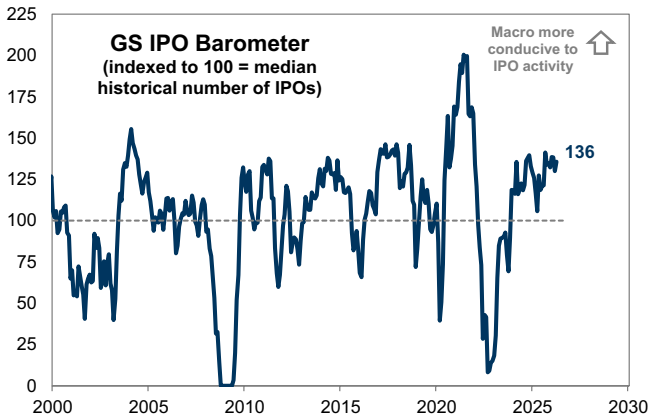
Exhibit 35: S&P 500 implied volatility



Source: Goldman Sachs Global Investment Research

IPO Barometer and mutual fund performance

Exhibit 36: Goldman Sachs IPO Barometer



The Goldman Sachs IPO Barometer combines five indicators that gauge the conduciveness of the macro environment to IPO activity.

Source: Goldman Sachs Global Investment Research

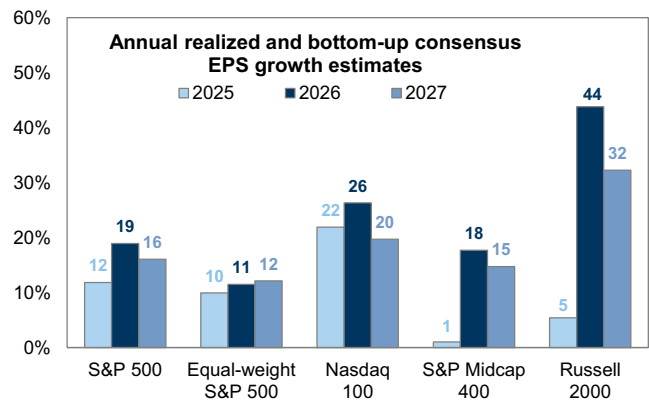
Exhibit 37: US equity mutual fund returns relative to benchmarks

Mutual fund style	Average	% outperforming
	YTD return	benchmarks YTD
Large-cap core	4%	49%
Large-cap growth	1	63
Large-cap value	6	21
Large-cap total	4%	45%
Small-cap core	11%	33%

Source: FactSet, Goldman Sachs Global Investment Research

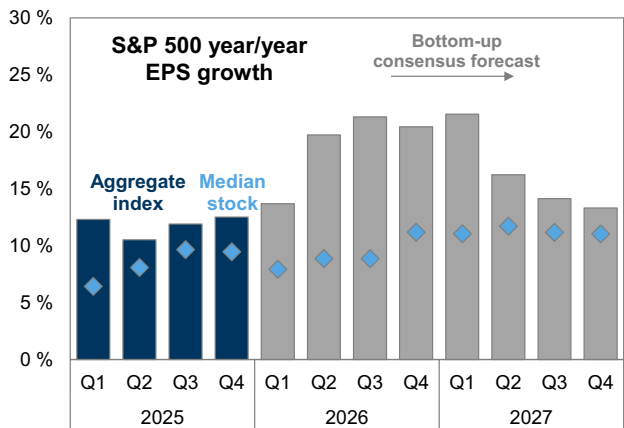
Earnings growth

Exhibit 38: Realized and consensus EPS growth for select US equity indices



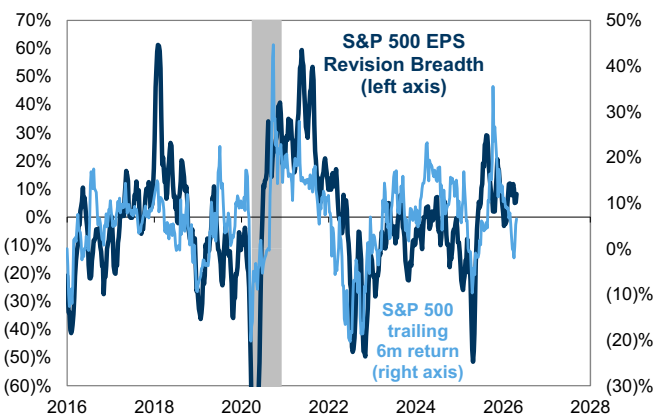
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 39: Realized and consensus year/year S&P 500 EPS growth



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 40: S&P 500 FY2 earnings revision breadth



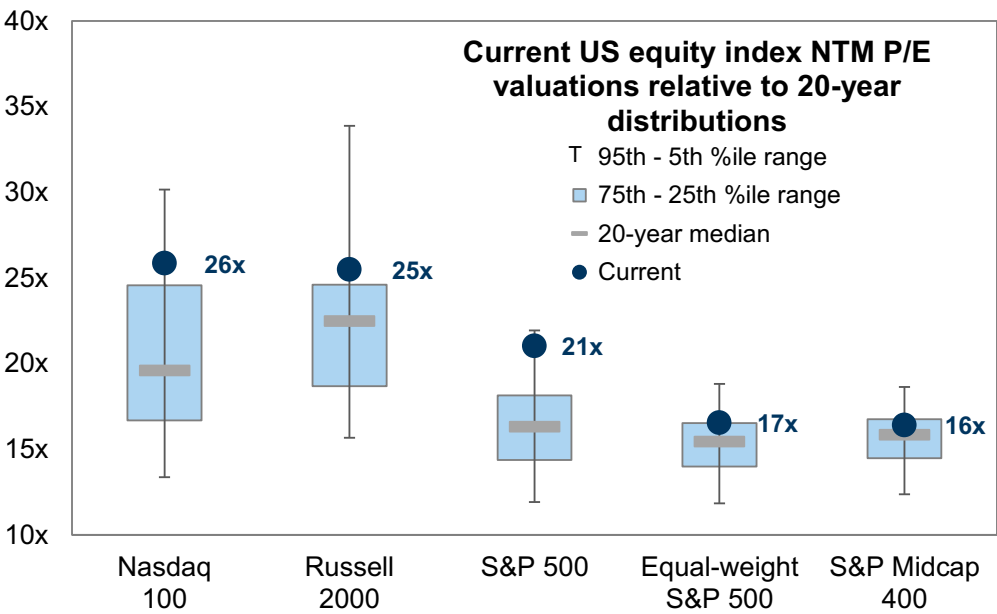
Revision breadth calculated as $\frac{[(\text{\# of companies with positive EPS revisions}) - (\text{\# of companies with negative revisions})]}{(\text{total \# of companies})}$. Revisions reflect FY2 EPS estimate changes during a rolling 1-month window.

Source: FactSet, Goldman Sachs Global Investment Research

For the exclusive use of SASCHA.SHA@SINOPAC.COM

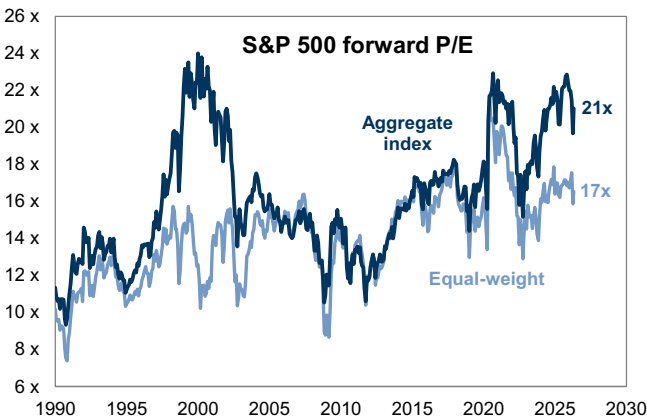
Valuations

Exhibit 41: US equity index P/E valuations vs. history



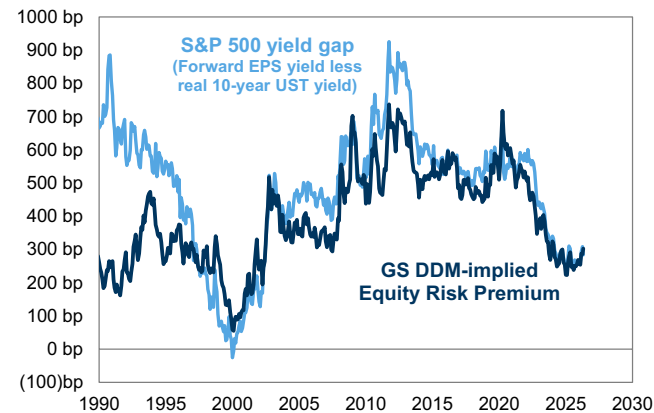
Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Exhibit 42: S&P 500 consensus forward 12-month P/E



Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Exhibit 43: S&P 500 valuation relative to US Treasury yields



Equity risk premium calculated by solving for the discount rate that equates the fair value of our multi-stage S&P 500 dividend discount model to the current market price and subtracting the 10-year US Treasury yield.

Source: Goldman Sachs Global Investment Research

For the exclusive use of SASCHA.SHA@SINOPAC.COM

Sector returns, earnings, and valuations

Exhibit 44: Sector and industry group returns

			Top quartile						
			Bottom quartile						
S&P 500			Index weight	1 Week	1 Month	3 Months	Last 12 Mo	YTD	GS allocation
			100 %	1.0 %	8 %	3 %	34 %	4 %	
S E C T O R	Consumer Staples	5 %	3.0 %	4 %	3 %	7 %	10 %	OW OW	
	Information Technology	35	2.2	13	7	56	6		
	Industrials	9	2.2	7	6	40	13		
	Energy	3	0.4	(5)	16	45	28		
	Real Estate	2	0.4	9	11	16	13		
	Consumer Discretionary	10	0.3	9	(3)	28	0		
	Materials	2	(0.1)	9	3	29	14		
	Health Care	9	(0.3)	1	(7)	10	(5)		
	Utilities	2	(0.5)	3	9	21	9		
	Financials	12	(0.5)	5	(2)	10	(5)		
Communication Services		11	(0.9)	9	3	53	5		
I N D U S T R Y G R O U P	Consumer Staples Retail	2 %	4.2 %	6 %	7 %	22 %	17 %		
	Technology Hardware & Equipment	10	3.8	12	17	53	11		
	Semiconductors & Semi Equipment	16	3.4	21	16	125	21		
	Transportation	1	2.9	9	8	35	10		
	Food Beverage & Tobacco	2	2.5	3	2	5	8		
	Capital Goods	7	2.3	7	9	55	17		
	Health Care Equipment & Services	3	2.0	4	(8)	(6)	(8)		
	Consumer Discretionary Retail	6	1.9	16	2	30	8		
	Telecommunication Services	1	1.7	(3)	14	2	11		
	Consumer Durables & Apparel	0	1.5	2	(5)	10	(3)		
	Household & Personal Products	1	1.2	0	(4)	(11)	2		
	Energy	3	0.4	(5)	16	45	28		
	Real Estate Investment Trusts (REITs)	2	0.4	9	11	16	13		
	Banks	3	0.2	9	3	40	(1)		
	Commercial & Professional Svcs	1	0.0	(2)	(12)	(20)	(11)		
	Materials	2	(0.1)	9	3	29	14		
	Insurance	2	(0.2)	3	2	(3)	(2)		
	Utilities	2	(0.5)	3	9	21	9		
	Financial Services	7	(0.9)	4	(5)	3	(7)		
	Consumer Services	2	(1.0)	3	(3)	10	(2)		
	Media & Entertainment	10	(1.1)	10	2	60	4		
	Pharma Biotech & Life Sciences	5	(1.6)	(0)	(6)	22	(4)		
	Software & Services	9	(1.6)	3	(13)	1	(18)		
	Automobiles & Components	2	(3.4)	(1)	(16)	49	(16)		

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 45: Earnings contributions and growth by S&P 500 sector

Sector	GS top-down					Consensus bottom-up			
	Contribution			EPS growth		Contribution		EPS growth	
	2025	2026	2027	2026	2027	2026	2027	2026	2027
Info Tech	\$70	\$92	\$109	31 %	19 %	\$101	\$127	44 %	25 %
Financials	52	55	59	6	6	56	62	8	11
Health Care	33	35	37	5	6	35	39	4	13
Comm Services	29	31	35	7	13	33	38	13	15
Cons Discretionary	23	23	25	4	5	24	28	7	15
Industrials	21	23	25	7	9	23	27	9	18
Consumer Staples	15	15	16	3	4	15	17	4	8
Energy	12	13	13	10	2	17	16	41	(1)
Utilities	8	8	9	6	7	9	9	13	9
Materials	5	7	7	24	8	7	8	35	10
Real Estate	6	7	7	5	7	7	8	7	16
S&P 500	\$275	\$309	\$342	12 %	10 %	\$327	\$379	19 %	16 %

Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 46: S&P 500 sector P/E valuations relative to history

	Consensus forward 12-month P/E valuation				
	Current P/E	Absolute P/E		P/E vs. S&P 500	
		%ile rank vs. history		%ile rank vs. history	
		10-year	30-year	10-year	30-year
Consumer Discretionary	28x	66%	88%	45%	79%
Industrials	26	94	98	100	100
Information Technology	23	56	63	54	42
Consumer Staples	22	87	87	53	31
Communication Services	22	76	77	45	18
S&P 500	21	64	81		
Utilities	19	81	94	80	81
Materials	19	61	77	32	20
Real Estate	18	55	72	32	30
Health Care	17	59	51	26	12
Energy	15	42	59	46	25
Financials	15	41	67	4	6

Source: Compustat, FactSet, IBES, Goldman Sachs Global Investment Research

Thematic baskets

Exhibit 47: Recent performance of select Goldman Sachs Research baskets

basket returns expressed relative to the equal-weight S&P 500 except for long/short basket pairs as noted

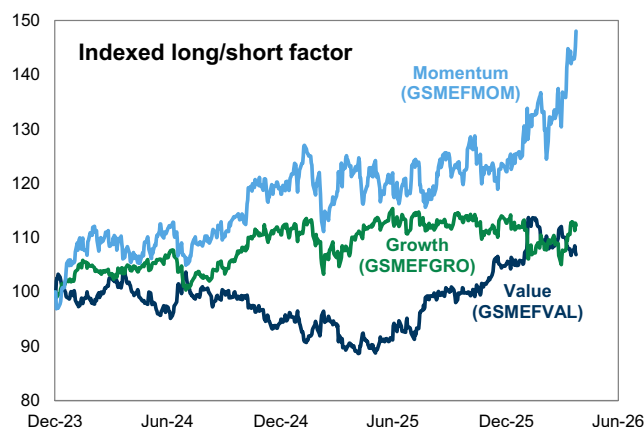
	Basket / Index	Ticker (GSTH_____)	Total returns			
			1 Wk	3 Mo	12 Mo	YTD
					Top quartile	Bottom quartile
	S&P 500	SPX	1.0 %	3 %	34 %	4 %
	Equal-weight S&P 500	SPW	0.9	2	25	6
Macro-economic	Dual Beta	BETA	2.9 %	14 %	84 %	23 %
	Interest Rate Sensitive	USTY	2.4	8	27	9
	Low vs. High Labor Cost	LLAB / HLAB	2.1	11	23	13
	High vs. Low Tax Rate	HTAX / LTAX	(0.9)	(1)	(4)	2
Geographic sales	International Sales	INTL	3.4 %	14 %	43 %	20 %
	EM Sales	BRIC	3.4	19	66	29
	US Sales	AINT	1.9	(6)	(21)	(10)
	Western Europe Sales	WEUR	1.0	1	11	2
Fundamental	ROE Growth	GROE	4.4 %	19 %	42 %	24 %
	Strong vs. Weak Balance Sheet	SBAL / WBAL	3.0	(7)	(20)	(7)
	High Quality	QUAL	0.2	(4)	(9)	(4)
	Long vs. Short Duration	LDUR / SDUR	(0.1)	(23)	(30)	(24)
	High vs. Low Operating Leverage	OPHI / OPLO	(1.0)	(6)	(14)	(12)
	Stable Growers	STGR	(1.5)	(5)	(18)	(6)
Uses of cash	High Growth Investment	HGIR	3.1 %	10 %	36 %	12 %
	Capex and R&D	CAPX	1.8	14	47	15
	Buybacks	REPO	1.0	1	3	(3)
	Dividend Growth	DIVG	1.0	2	10	(0)
	Debt Reducers vs. Issuers	DRED / DISS	0.7	7	27	9
	Total Cash Return	CASH	0.3	3	10	(2)
Ownership, risk & liquidity	Mutual Fund Over- vs. Underweights	MFOV / MFUW	1.5 %	(6)%	(23)%	(11)%
	Low vs. High Liquidity	LLIQ / HLIQ	0.0	(1)	(5)	4
	Hedge Fund VIP	HVIP	(0.6)	1	19	2
	High Sharpe Ratio	SHRP	(1.0)	(10)	(17)	(11)
Equity research	SMB Exposure	GSRHSMBX	0.9 %	(9)%	(17)%	(13)%
	Oil Input Cost	GSRHOILX	(2.2)	(4)	(18)	(0)
	M&A Candidates	GSRHACQN	(2.4)	(6)	23	(8)
	Gov't Exposure	GSRHGOVT	(3.3)	(5)	25	(1)

Basket returns and constituents can be accessed via GS Marquee or Bloomberg using the indicated tickers.

Source: Goldman Sachs Global Investment Research

Factors

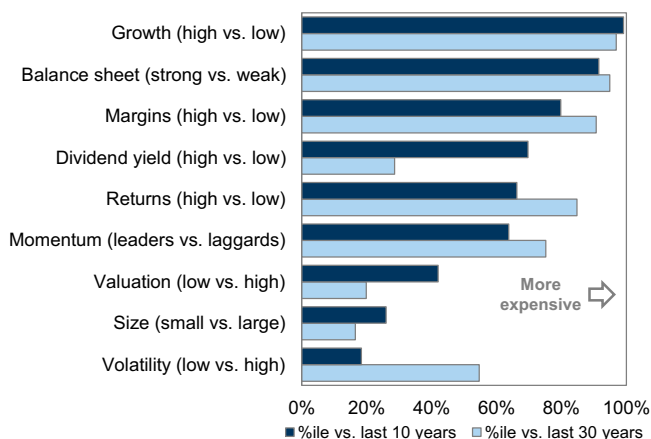
Exhibit 48: Indexed return of select equity factors



Factor performance and constituents can be viewed on GS Marquee or Bloomberg using the tickers indicated on the chart.

Source: Goldman Sachs Global Investment Research

Exhibit 49: Equity factor valuations relative to history



Source: Goldman Sachs Global Investment Research

Goldman Sachs global macro research cross-asset forecasts

Exhibit 50: Goldman Sachs macro research asset forecasts

Goldman Sachs Global Macro Forecasts

	units	Current	3m	6m	12m	Change to 12m target
Equities						
TOPIX	level	3716	3800	4000	4200	13 %
MXAPJ	level	819	830	870	920	12
S&P 500	level	7108	7200	7400	7600	7
STOXX Europe 600	level	614	605	615	625	2
10-year rates						
Euro Area (Germany)	%	2.5	2.9	3.0	3.0	55 bp
US	%	4.3	4.2	4.1	4.1	(22)
Japan	%	2.4	2.0	2.0	2.0	(42)
Corporate bonds						
High yield	bp	272	335	325	NA	NA
Investment grade	bp	80	95	92	NA	NA
Currencies						
Euro / US dollar	EUR/\$	1.17	1.14	1.18	1.20	2 %
Sterling / US dollar	£/\$	1.35	1.33	1.34	1.33	(1)
US dollar / Yen	\$/¥	159	160	158	155	(3)
Commodities						
NYMEX nat. gas	\$/mmBtu	3	3.50	3.50	3.50	34 %
COMEX gold	\$/troy oz	4724	4805	4990	5445	15
LME copper	\$/mt	13190	12635	12535	12265	(7)
Brent crude oil	\$/bbl	105	84	80	80	(24)

Source: FactSet, Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Ben Snider, Ryan Hammond, Jenny Ma, Daniel Chavez, Kartik Jayachandran and Christophe Sung, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Ben Snider Goldman Sachs & Co. LLC, Ryan Hammond Goldman Sachs & Co. LLC, Jenny Ma Goldman Sachs & Co. LLC, Daniel Chavez Goldman Sachs & Co. LLC, Kartik Jayachandran Goldman Sachs & Co. LLC, Christophe Sung Goldman Sachs & Co. LLC.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Basket disclosure

The ability to trade the basket(s) in this report will depend upon market conditions, including liquidity and borrow constraints at the time of trade.

Marquee disclosure

Marquee is a product of Goldman Sachs FICC and Equities. Any Marquee content linked in this report is not necessarily representative of GS Research views. Because the Marquee platform and third-party platforms (e.g., Bloomberg) are only permitted to reflect changes to these baskets after publication of this note, there will be a delay between the rebalancing of baskets in this note and when the changes to these baskets are reflected on such platforms. If you need access to Marquee, please contact your GS salesperson or email the Marquee team at gs-marquee-sales@gs.com.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: <https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf>, and a copy of the annual audit compliance report can be found at this link: <https://publishing.gs.com/content/site/india-annual-compliance-report.html>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs

Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>.

Russia: Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate.

Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.